

Exploring What Drives an Improvement Mindset Among Physicians

Shirling Liu^{1,2}, Laura Desveaux^{2,3,4}, Ruoxi Wang², Simona C. Minotti^{2,1}

¹ Dalla Lana School of Public Health, Biostatistics Division, University of Toronto, Toronto, Canada.

² Institute for Better Health, Trillium Health Partners, Mississauga, Canada.

³ Institute of Health Policy, Management and Evaluation, University of Toronto, Toronto, Canada.

⁴ Women's College Hospital Research Institute, Women's College Hospital, Toronto, Canada.

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Abstract

Background: Healthcare systems require continuous learning and adaptation to address challenges from aging populations to workforce burnout, yet physician engagement with performance feedback remains suboptimal. Data-driven performance feedback is widely used to support practice changes and highlight the importance of understanding the cognitive beliefs underlying an improvement mindset among physicians.

Objectives: This study aims to identify latent subgroups of primary care physicians' (PCPs) beliefs and assess whether subgroup membership predicts perceived accountability to act on feedback.

Methods: Using a cross-sectional design, this study evaluates survey data from 206 Ontario PCPs via validated psychometric scales. Confirmatory Factor Analysis (CFA) was utilized to

validate the underlying structure (constructs) of the Feedback Orientation Scales (FOS) and Organizational Change Recipients' Beliefs Scale (OCRBS), and inputting these validated factors into Latent Profile Analysis (LPA) to identify latent subgroups of physician belief profiles, and fitting regression model to evaluate how subgroup membership predicts feedback accountability.

Results: Physicians demonstrated a strong baseline consensus on the need for clinical change, with the positive correlations of the five cognitive constructs: Feedback Utility, Feedback Accountability, Feedback Self-Efficacy, Valence, and Change Discrepancy. CFA validated this 5-factor structural model and later identified two distinct physician subgroups by LPA based on four cognitive constructs (excluding Accountability): a “High-Appraisal” ($n = 28$) and a “Low-Appraisal” ($n = 178$) profile. Multivariable regression confirmed High-Appraisal membership positively predicted Feedback Accountability, independent of covariates like gender, international graduate status, and workdays per week.

Conclusions: The findings confirm that even after controlling for the effects of baseline demographics and practice characteristics covariates, physician accountability for feedback is influenced by underlying cognitive appraisals. Identifying specific belief patterns provides a detailed understanding necessary for future interventions.

Introduction

Background

Health care systems require continuous learning to address challenges from aging populations to workforce burnout. Although there are customized audit-and-feedback interventions for specific clinical settings, the challenge of variability in physicians’ engagement with and response to the feedback still remains. This variability determines the effectiveness of the intervention and serves as a strong rationale to examine the individual cognitive appraisals behind this engagement. Supporting engagement with data-driven feedback requires understanding the cognitive beliefs underlying an improvement mindset [1]. While previous research [1] has mapped population-level variable interactions, this study proposes a conceptual framework (Figure 1) using a person-centred statistical approach [2] to identify distinct latent physician subgroups.

Theoretical Framework

This study utilizes two validated psychometric frameworks:

- **Feedback Orientation Scale (FOS)** [3]: Evaluates an individual’s cognitive receptivity to feedback across five dimensions. This study involved three of the FOS domains: *Feedback Utility* (the perceived usefulness of the feedback), *Feedback Self-Efficacy* (the physician’s confidence in their ability to act on it), and *Feedback Accountability* (the physician’s sense of accountability to react to feedback).
- **Organizational Change Recipients’ Beliefs Scale (OCRBS)** [4, 5]: Evaluates an individual’s readiness for change across five criterion variables. This study investigated two of the OCRBS domains: *Change Discrepancy* (recognizing the gap between current performance and desired clinical standards) and *Valence* (the perceived value of the feedback in aligning with clinical practice discrepancies).

Based on control-value theory, *Feedback Accountability* is treated as a downstream motivational response in this study, whereas the other four constructs (*Utility*, *Self-Efficacy*, *Valence*, and *Discrepancy*) represent the antecedent cognitive appraisals that shape that response [1, 6, 7]. For this reason, *Accountability* serves as the primary outcome variable to be predicted, while the remaining four appraisal constructs are utilized to derive the latent cognitive profiles.

By integrating these five cognitive constructs from the FOS and OCRBS, this study evaluates both the internal mindset and external evaluations of physicians regarding current clinical practice. Before analyzing, the priority of this project is to deeply understand the psychological concepts of these five constructs. Crucially, “*Valence*” and “*Utility*” represent two distinct ways physicians process information. For instance, a physician might perceive feedback as unaligned with the clinical practice discrepancy (negative valence) but still recognize high utility if the data provides an actionable way for their clinical improvement, and vice versa. This distinction clarifies that *Valence* (affective endorsement) and *Utility* (actionability) are independent constructs.

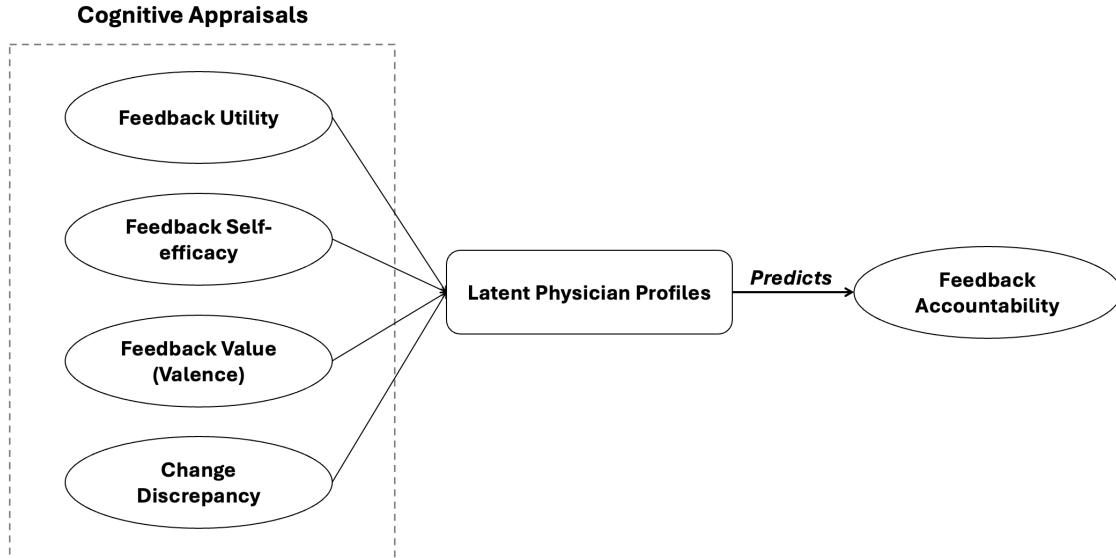


Figure 1: Person-centered conceptual model integrating FOS and OCRBS cognitive appraisals to derive distinct latent physician profiles that predict feedback accountability.

Study Objectives

This study integrates these frameworks to capture how primary care physicians (PCPs) process performance data. The primary objectives are to:

1. Identify latent subgroups of primary care physicians' (PCPs) belief patterns through validated factor structures; and
2. Assess whether subgroup membership predicts perceived accountability to act on feedback.

Materials and Methods

Study Design and Target Population

This study utilized cross-sectional survey data of registered primary care physicians (PCPs) in Ontario ($N = 1,251$). The survey instruments measured a total of five cognitive constructs. Specifically, the FOS was used to assess three cognitive appraisals using a 5-item scale for each construct, while the OCRBS assessed the remaining two cognitive appraisals using a 4-item scale for each construct. Responses were recorded using a 5-point Likert scale (ranging from *Strongly Disagree* to *Strongly Agree*), alongside standard demographic and practice characteristics, such as gender and years in practice.

Variables and Data Preprocessing

- **Primary Outcome Variable:** Continuous factor score for *Feedback Accountability*.
- **Primary Predictor Variable:** Categorical *Profile* assignment derived from the LPA.
- **Secondary Independent Variables:** Baseline demographic and practice characteristics including gender, international graduate status, years in practice, workdays per week, practice model, community size, and number of staff.

After data quality checks, responses generated by bots and observations that were completely missing from the original dataset were excluded, and the final analysis sample consisted of 206 physicians (see Figure 2). The remaining missing data and ‘Prefer not to answer’ were standardized as **NA**. Questionnaire items were standardized to an ordinal 5-point Likert scale ranging from Strongly Disagree (1) to Strongly Agree (5). Small categorical demographics were collapsed into binary format to ensure adequate statistical power and satisfy regression assumptions (i.e., years in practice combined as “5 years or less” or “6 years or more”). Exploratory data analysis of raw survey items revealed a strong baseline consensus with high clustering and negative skewness, and median scores primarily of 4 and a few of 3. Cronbach’s α ranged from 0.71 to 0.86, indicating that all items within each construct demonstrated strong internal consistency.

Statistical Modelling Pipeline

Analyses were executed in R (version 4.5.3) following a three-step sequential pipeline:

1. **Confirmatory Factor Analysis (CFA):** CFA was conducted using the `lavaan` package in R. To validate the ordinal items of the five-factor survey construct, CFA was estimated using the weighted least squares mean and variance adjusted (WLSMV) estimator with item-level polychoric correlations to account for the ordinal nature of the items. Likelihood ratio tests (scaled chi-square difference tests) appropriate for the WLSMV estimator were used to make statistical comparisons between nested models. Model fit was evaluated using standard thresholds: Comparative Fit Index (CFI) and Tucker-Lewis Index (TLI) > 0.95 [8], and Root Mean Square Error of Approximation (RMSEA) < 0.08 [9]. The measurement model is given by:

$$X_i = \lambda_i \xi + \delta_i$$

where X_i is the observed response for item i , ξ is the latent cognitive construct, λ_i is

the factor loading, and δ_i is the item-specific measurement error.

2. **Latent Profile Analysis (LPA)** [10]: LPA was conducted using the `tidyLPA` package in R. Categorical latent subgroups were extracted based on the continuous factor scores derived from the CFA. Candidate models ranging from 1 to 6 profiles were iteratively evaluated for Models 1, 2, 3, and 6. Optimal model selection followed Ram and Grimm (2009) [11], prioritizing theoretical plausibility, minimizing the Bayesian and Akaike Information Criteria (BIC/AIC), maximizing entropy, and obtaining a significant Bootstrapped Likelihood Ratio Test (BLRT) p -value. The mixture model is defined as:

$$f(y_l) = \sum_{k=1}^K \pi_k f_k(y_l \mid \mu_k, \Sigma_k)$$

Where $f(y_l)$ is the overall probability density of scores for individual l across K latent profiles. For each profile k , π_k is the probability of membership, and f_k is the profile-specific density function defined by its mean vector (μ_k) and covariance matrix (Σ_k).

3. **Multiple Linear Regression (MLR)**: MLR was utilized to analyze whether categorical physician belief profile membership derived from LPA predicts Feedback Accountability. AIC and BIC bidirectional stepwise selection were used to identify the most parsimonious model from the full covariate set. The regression model is defined as:

$$Y_{\text{Accountability}} = \beta_0 + \beta_1 \text{Profile}_2 + \cdots + \beta_{k-1} \text{Profile}_k + \epsilon$$

where $Y_{\text{Accountability}}$ is the predicted score, β_0 is the baseline profile's mean, the β coefficients are the score differences for other profiles, Profile are binary membership indicators, and ϵ is the residual error.

Model validation was conducted on the final model to assess multicollinearity using Variance Inflation Factors (VIF), visually evaluate normality via Q-Q plots, assess homoscedasticity via residuals-versus-fitted plots, and examine influential observations

via Cook's distance thresholds.

Results

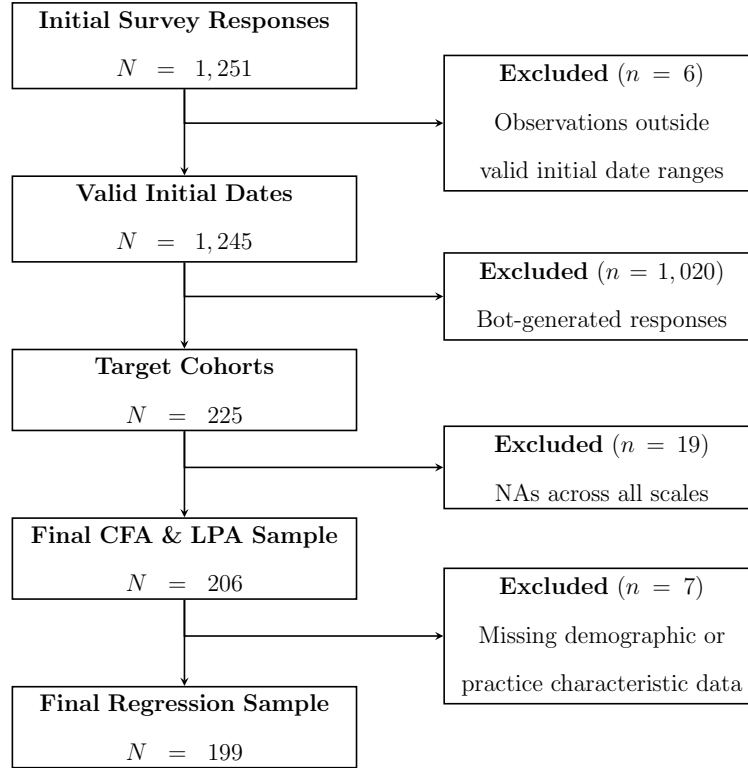


Figure 2: Participant flow diagram detailing the sequential exclusion criteria from initial survey responses to the final analytical samples.

Factor Validation (CFA)

The CFA model demonstrated excellent fit for the hypothesized 5-factor structure across all standard indices (see Table 1) (CFI = 0.976, TLI = 0.972, RMSEA = 0.051, SRMR = 0.062). Distinctiveness and discriminant validity of the construct were confirmed using a scaled chi-square difference test, which showed better fit of the 5-factor model (scaled $\chi^2 = 335.41$, $df = 220$, $p < .001$) relative to other possible simpler alternatives such as 4-factor (merging Utility/Valence) or 2-factor (FOS vs. OCRBS) models.

Table 1: Confirmatory Factor Analysis (CFA) Fit Indices Comparing the 5-Factor Measurement Model Against Competing Alternatives ($N = 206$).

Model	Absolute Fit			Incremental Fit				Residual Fit	
	χ^2 (scaled)	df	p	CFI	TLI	Robust CFI	Robust TLI	RMSEA	SRMR
5-Factor (Hypothesized)	335.41	220	< .001	0.976	0.972	0.890	0.874	0.051	0.062
4-Factor (Merge Utility/Valence)	399.33	224	< .001	0.963	0.958	0.858	0.840	0.062	0.070
4-Factor (Merge Utility/Accountability)	358.94	224	< .001	0.971	0.968	0.880	0.864	0.054	0.065
2-Factor (FOS vs OCRBS)	909.04	229	< .001	0.856	0.841	0.669	0.635	0.120	0.120
1-Factor (Global)	1234.77	230	< .001	0.788	0.766	0.576	0.533	0.146	0.140

While the robust CFI (0.890) and TLI (0.874) are somewhat lower than the commonly used guidelines, they reflect the fact that the mean- and variance-adjusted scaling corrections applied in WLSMV estimation, as these scaled fit indices have been shown to converge to distorted asymptotic values [12]. Standardized factor loadings (λ) were statistically significant, ranging from 0.49 to 0.89 (all $p < .001$), indicating stable measurement of the latent belief system. Latent correlations revealed that Accountability is strongly associated with both Feedback Utility ($r = 0.86$) and Valence ($r = 0.71$) (Figure 3).

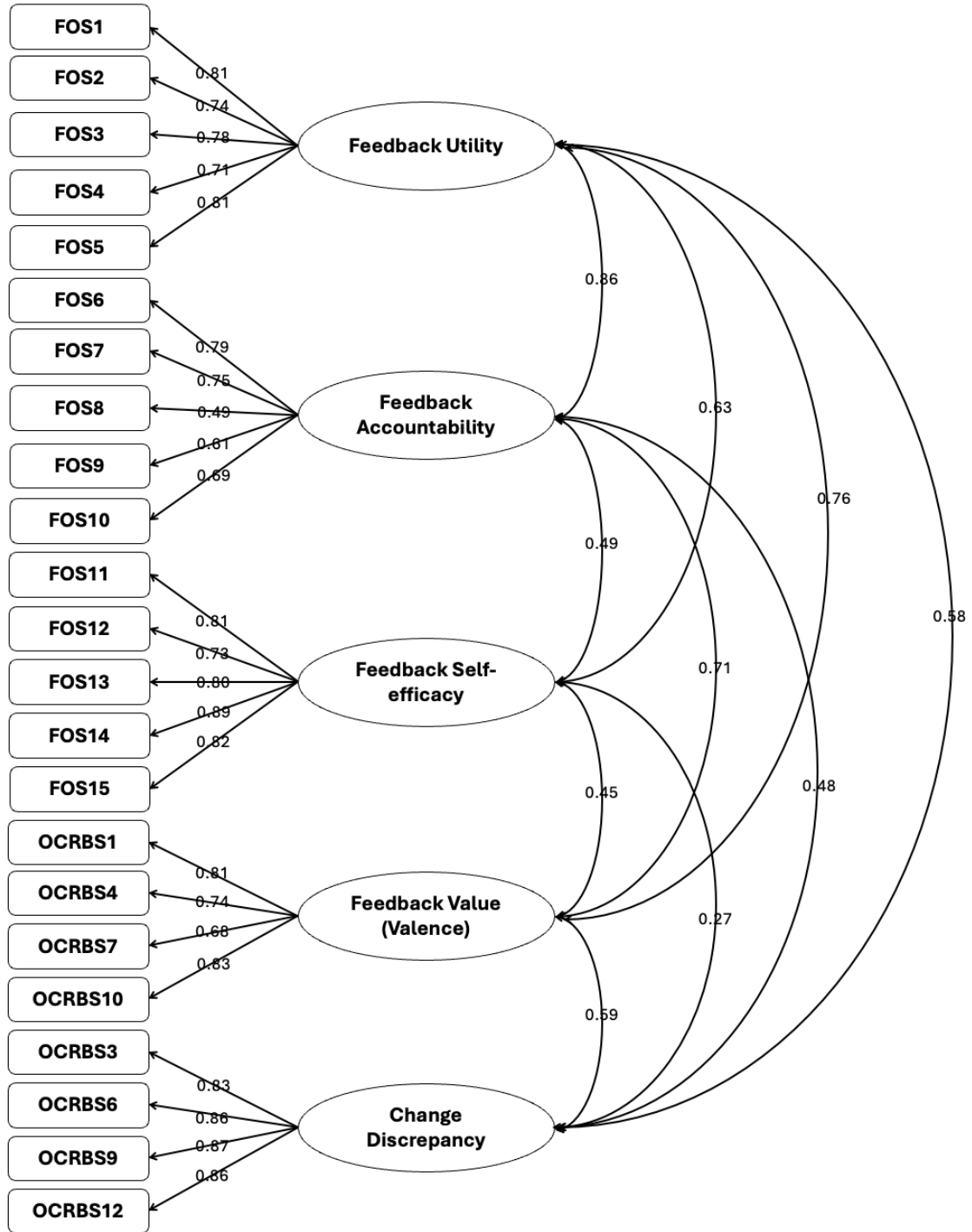


Figure 3: Confirmatory Factor Analysis (CFA) path diagram showing standardized factor loadings (λ) and latent correlations. (Note: All factor loadings and latent correlations are statistically significant at $p < .001$.)

Latent Profile Analysis (LPA)

Based on the validated continuous latent factor scores for Utility, Self-Efficacy, Valence, and Discrepancy, LPA was conducted to identify latent subgroups (see Table 5 in Appendix A). Model selection followed the decision framework proposed by Ram and Grimm (2009) [11] and was supported by the automated `compare_solutions()` function. The final comparison focused on the 3-Class Model 1, 2-Class Model 3, and 4-Class Model 6 solutions (see Table 2).

Table 2: Comparison of Candidate Latent Profile Models: Fit Indices and Class Size Characteristics.

Model	Parameters	Log-Likelihood	AIC	BIC	SABIC	Entropy	Minimum n (%)	BLRT <i>p</i> -value
3-Class (Model 1)	18	-729.75	1495.51	1555.41	1498.38	0.86	15.0%	0.010
2-Class (Model 3)	25	-645.38	1340.77	1423.96	1344.75	0.82	13.6%	0.010
4-Class (Model 6)	83	-532.55	1231.10	1507.31	1244.34	0.91	9.7%	0.010

Note: AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion; SABIC = Sample-Size Adjusted BIC; BLRT = Bootstrapped Likelihood Ratio Test.

Even though the `compare_solutions()` function provided the best-fitting 4-Class Model 6 solution based on fit indices, the model had 83 estimated parameters with a sample size of 206, which raises concerns about the complexity of the model and potential overfitting. Also, the potential estimation instability was evidenced by the non-sequential BLRT pattern in the Model 6 sequence. Therefore, BIC was prioritized as the primary model selection criterion, while also considering AIC, entropy, minimum class size, and model parsimony. The 2-Class Model 3 solution was selected as the final model.

The two extracted subgroups were classified as (Figure 4):

- **High-Appraisal** ($n = 28$): Characterized by above-average continuous scores across all cognitive domains.
- **Low-Appraisal** ($n = 178$): Characterized by consistently below-average scores across the constructs.

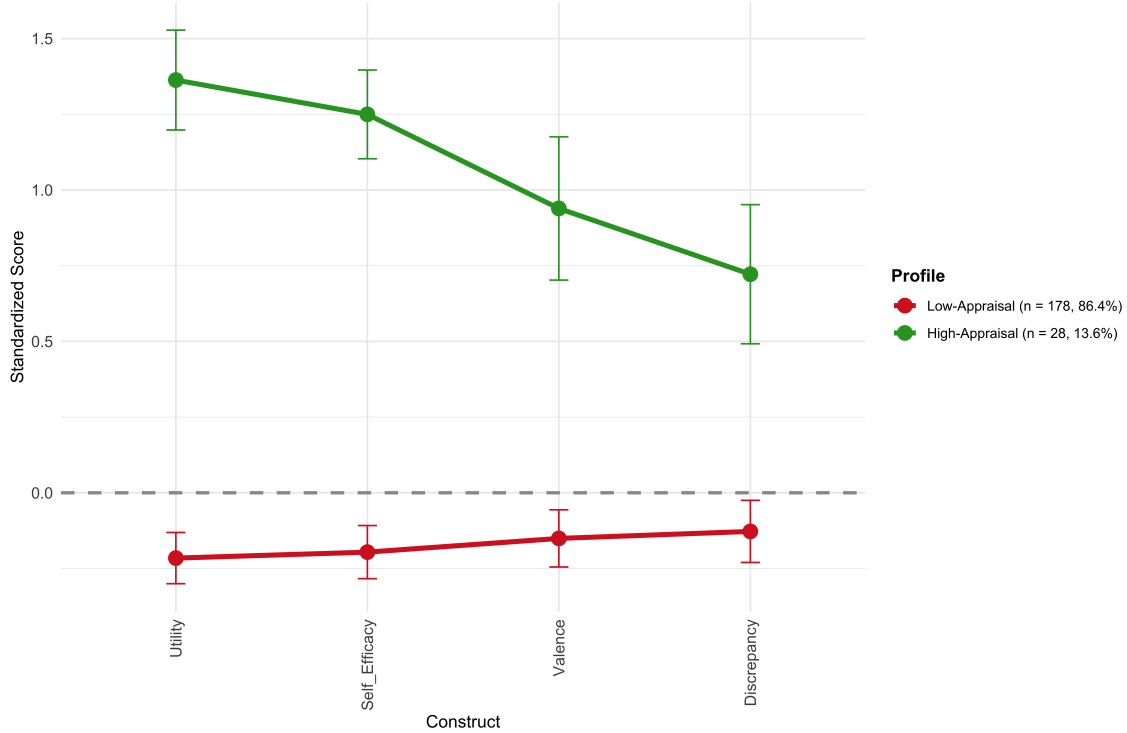


Figure 4: Standardized Mean Scores by Latent Profile. The dashed line represents the overall sample average.

Predictors of Accountability

The final model, excluding 7 missing observations ($N = 199$) (see Table 6 in Appendix A), using multiple linear regression was selected using bidirectional selection based on the AIC criterion (see Table 3), which included the latent belief profile along with gender, international graduate status, and workdays per week. Validation of the model showed that the multicollinearity assumption was met, where the Variance Inflation Factor (VIF) values were below the conservative threshold of 5.0 (VIF ranged from 1.01 to 1.02). Residual analysis verified that linearity, normality, and homoscedasticity assumptions were met, without any highly influential observations crossing Cook's distance criteria.

Table 3: Comparison of Regression Models (2-Class Model 3): Evaluating Variable Retention and Model Fit across Selection Criteria

Characteristic	Full Model			Stepwise (AIC) / Final			Stepwise (BIC)		
	Beta	95% CI	p-value	Beta	95% CI	p-value	Beta	95% CI	p-value
(Intercept)	-0.28	-0.56, 0.00	0.048	-0.21	-0.38, -0.04	0.014	-0.18	-0.26, -0.09	< 0.001
Physician Belief Profiles			< 0.001			< 0.001			< 0.001
<i>Low-Appraisal</i>	—	—		—	—		—	—	
<i>High-Appraisal</i>	1.4	1.1, 1.6	< 0.001	1.4	1.1, 1.6	< 0.001	1.3	1.1, 1.6	< 0.001
Gender			0.042			0.048			
<i>Man</i>	—	—		—	—				
<i>Woman</i>	0.17	0.01, 0.34	0.042	0.16	0.00, 0.32	0.048			
International Graduate			0.106			0.131			
<i>No</i>	—	—		—	—				
<i>Yes</i>	0.18	-0.04, 0.40	0.106	0.16	-0.05, 0.37	0.131			
Years in Practice			0.444						
<i>5 years or less</i>	—	—							
<i>6 years or more</i>	0.07	-0.11, 0.25	0.444						
Work Days per Week			0.091			0.068			
<i>3 or fewer days per week</i>	—	—		—	—				
<i>4 or more days per week</i>	-0.15	-0.32, 0.02	0.091	-0.15	-0.31, 0.01	0.068			
Practice Community Size			0.588						
<i>Urban centre</i>	—	—							
<i>Smaller areas</i>	-0.04	-0.21, 0.12	0.588						
Number of Staff			0.621						
<i>5 or less</i>	—	—							
<i>6 or more</i>	0.04	-0.13, 0.21	0.621						
Practice Model			0.822						
<i>FHO</i>	—	—							
<i>Other</i>	0.02	-0.14, 0.18	0.822						
R^2		0.438			0.435			0.409	
Adjusted R^2		0.414			0.423			0.406	
Statistic		18.5			37.3			136	
p-value		< 0.001			< 0.001			< 0.001	
AIC		344			337			340	
BIC		376			356			349	
Total Physicians (N)		199			199			199	

Abbreviation: CI = Confidence Interval

The final model (see Table 4) explained approximately 42.3% of the variance in accountability scores (Adjusted $R^2 = 0.423$). Profile membership emerged as a highly significant predictor. The physicians in the High-Appraisal group demonstrated significantly higher accountability scores as compared to those in the Low-Appraisal baseline group ($\beta = 1.4$, $p < .001$). Of the demographic and practice characteristic variables, only gender emerged as a significant predictor in the final model ($\beta = 0.16$, $p < .001$).

Table 4: Predictors of Physician Accountability: Linear Regression Adjusted for Practice and Demographic Characteristics

Characteristic	Beta	95% CI	<i>p</i> -value
(Intercept)	-0.21	-0.38, -0.04	0.014
Physician Belief Profiles			< 0.001
<i>Low-Appraisal</i>	—	—	
<i>High-Appraisal</i>	1.4	1.1, 1.6	< 0.001
Gender			0.048
<i>Man</i>	—	—	
<i>Woman</i>	0.16	0.00, 0.32	0.048
International Graduate			0.131
<i>No</i>	—	—	
<i>Yes</i>	0.16	-0.05, 0.37	0.131
Work Days per Week			0.068
<i>3 or fewer days per week</i>	—	—	
<i>4 or more days per week</i>	-0.15	-0.31, 0.01	0.068

Abbreviations: CI = Confidence Interval.

Adjusted $R^2 = 0.423$; Total Physicians (N) = 199.

Discussion

The results of this study underscore that physician engagement with clinical performance data is associated with underlying cognitive appraisals. The CFA confirms that while physicians evaluate feedback through distinct cognitive constructs, the strong correlation between Feedback Utility, Valence, and Accountability clearly shows that there are interactions between these constructs.

Through the application of Latent Profile Analysis, the current study was able to transcend the population-level averages by defining two operational mindsets in physicians: high-appraisal and low-appraisal, distinguishing physicians by their overall positivity toward feedback rather than unique belief configurations (e.g., high valence with low utility). This cohesive clustering pattern of these appraisals aligns with control-value theory and demonstrates that physicians who consider feedback clearly recognize the discrepancy and value it, simultaneously perceive it as highly useful and feel strongly confident to act on it. This generalized mindset is consistent with previous research, which revealed that cognitive readiness and feedback orientation manifest as a unified tendency rather than as disparate and conflicting beliefs [1].

By applying regression analysis, the psychological profile was confirmed as the primary determinant of downstream accountability, independent of covariates. The results suggest that physicians with Low-Appraisal process their feedback fundamentally differently compared to High-Appraisal. Organizations should diagnose physicians' level of readiness and then develop strategies such as academic detailing and peer coaching.

Weaknesses and Limitations

The first limitation of this study is the use of a cross-sectional survey design. Though the regression analysis indicates strong predictive associations, it is unable to provide temporal causality between belief profiles and future accountability behaviour. Secondly, conducting

complete case analysis for missing values rather than imputation to respect the nature of the survey design introduces the risk of selection/non-response bias and loss of statistical power. Thirdly, relying on self-reports to measure the psychological variables involved the potential for social desirability bias, especially considering the negatively skewed consensus observed during exploratory data analysis. Lastly, the stability and generalizability of the profile-specific estimates might be affected due to the relatively small size of the High-Appraisal profile ($n = 28$), although it exceeds the standard minimum sample size recommended for LPA.

Future Directions

Future research could focus on longitudinal study designs. Implementing this dual psychometric framework within longitudinal, multi-indicator quality control trials will allow researchers to test true causality. Specifically, longitudinal tracking is required to determine whether targeted coaching interventions can successfully shift a physician from a Low-Appraisal profile to a High-Appraisal profile over time, thereby actively building a learning health system.

Conclusion

This study successfully integrated the FOS and OCRBS psychometric frameworks to map the cognitive drivers of an improvement mindset among primary care physicians. By validating a distinct 5-factor cognitive structure, this work enabled the identification of latent belief profiles that heavily predict feedback accountability. Regression analysis confirms that independent of baseline characteristics, physician engagement relies on these cognitive appraisals: recognizing physician engagement as a psychologically modifiable state allows organizations to design targeted support strategies that address the underlying variances in how physicians engage with feedback.

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Appendix

Appendix A: Additional Tables

Table 5: Fit Indices for All Estimated Latent Profile Analysis Solutions ($N = 206$)

Model	Classes	k	LogLik	AIC	BIC	SABIC	Entropy	Min Class (%)	BLRT p
Model 1: Equal variances, zero covariances									
	1	8	-935.86	1889.73	1916.35	1891.01	1.00	100.0	–
	2	13	-807.47	1640.94	1684.20	1643.01	0.84	26.2	< .01
	3	18	-729.75	1495.51	1555.41	1498.38	0.86	15.0	< .01
	4	23	-702.47	1450.94	1527.49	1454.61	0.94	6.8	< .01
	5	28	-670.99	1415.98	1509.18	1420.45	0.88	4.4	< .01
	6	33	-670.73	1407.46	1517.28	1412.72	0.84	4.4	0.030
Model 2: Varying variances, zero covariances									
	1	8	-935.86	1889.73	1916.35	1891.01	1.00	100.0	–
	2	17	-805.74	1645.48	1702.06	1648.19	0.83	27.2	< .01
	3	26	-723.01	1498.03	1584.55	1502.17	0.85	15.5	< .01
	4	35	-690.69	1451.38	1567.85	1456.96	0.84	13.1	< .01
	5	44	-669.24	1426.48	1572.90	1433.49	0.96	8.7	< .01
	6	53	-653.16	1412.32	1588.69	1420.77	0.84	5.8	0.020
Model 3: Equal variances, equal covariances									
	1	20	-656.24	1352.47	1419.03	1355.66	1.00	100.0	–
	2*	25	-645.38	1340.77	1423.96	1344.75	0.82	13.6	< .01
	3	30	-639.71	1339.43	1439.27	1344.21	0.79	13.1	0.158
	4	35	-631.35	1332.71	1449.19	1338.30	0.72	12.6	0.099
	5	40	-626.45	1332.91	1466.02	1339.29	0.75	5.3	0.248
	6	45	-618.93	1327.87	1477.62	1335.04	0.79	6.3	0.069
Model 6: Varying variances, varying covariances									
	1	20	-656.24	1352.47	1419.03	1355.66	1.00	100.0	–
	2	41	-633.15	1348.30	1484.75	1354.84	0.66	25.7	0.040
	3	62	-619.12	1362.23	1568.56	1372.12	0.93	14.6	0.495
	4	83	-532.55	1231.10	1507.31	1244.34	0.91	9.7	< .01
	5	104	-559.58	1325.17	1671.27	1341.76	0.96	8.3	0.901
	6	125	-534.34	1318.68	1734.57	1338.62	0.94	8.3	0.020

Note. k = # of estimated parameters; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion; SABIC = Sample-Size Adjusted BIC; BLRT = Bootstrapped Likelihood Ratio Test.

* Selected final model (Model 3, 2-Class profile solution) prioritizing parsimony, BIC minimization, theoretical interpretability, and class distribution stability.

Table 6: Baseline Characteristics of the Final Sample

Characteristic	Overall ($N = 199$) ¹	Low-Appraisal ($N = 171$) ¹	High-Appraisal ($N = 28$) ¹
Gender			
Man	77 (39%)	63 (37%)	14 (50%)
Woman	122 (61%)	108 (63%)	14 (50%)
International Graduate	33 (17%)	27 (16%)	6 (21%)
Years in Practice			
5 years or less	60 (30%)	51 (30%)	9 (32%)
6 years or more	139 (70%)	120 (70%)	19 (68%)
Work Days per Week			
3 or fewer days per week	74 (37%)	64 (37%)	10 (36%)
4 or more days per week	125 (63%)	107 (63%)	18 (64%)
Practice Community Size			
Urban centre	94 (47%)	76 (44%)	18 (64%)
Smaller areas	105 (53%)	95 (56%)	10 (36%)
Number of Staff			
5 or less	91 (46%)	75 (44%)	16 (57%)
6 or more	108 (54%)	96 (56%)	12 (43%)
Practice Model			
FHO	92 (46%)	76 (44%)	16 (57%)
Other	107 (54%)	95 (56%)	12 (43%)

¹ n (%)

Appendix B: Program Syntax

Below is the complete R script used for data cleaning, descriptive statistics, Confirmatory Factor Analysis (CFA), Latent Profile Analysis (LPA), and final regression modeling.

```
# R Script: Data Cleaning, Descriptive Statistics, CFA, LPA, and Regression
```

```
### 01 Data Cleaning
```

```
library(tidyverse)
```

```
library(here)
```

```
library(lubridate)
```

```
library(skimr)
```

```
library(naniar)
```

```
library(table1)
```

```
## 1. Load Data
```

```
Aug_data <- read_csv(here("data", "raw", "IMPACT_survey_07Aug22 (original).csv"),
```

```

      locale = locale(encoding = "UTF-8"))
Dec_data <- read_csv(here("data", "raw", "IMPACT survey_23Dec22 (original).csv"),
                    skip = 2, col_names = names(Aug_data), locale = locale(encoding = "UTF-8"))
survey_data <- bind_rows(Aug_data, Dec_data)
dim(survey_data)
codebook <- read_csv(here("data", "raw", "IMPACT survey_codebook.csv"),
                    locale = locale(encoding = "UTF-8"))

## 2. Map and Combine Headers

# Create a mapping dataframe directly from the headers and first row
survey_mapping <- data.frame(
  question = colnames(survey_data),
  item = as.character(survey_data[1, ]),
  stringsAsFactors = FALSE
) %>% mutate(question_num = row_number())

# Add an question number to the codebook
codebook <- codebook %>% mutate(question_num = row_number())

# Match the codebook with survey_data by question number
mapped_data <- survey_mapping %>% left_join(codebook, by = "question_num")

# Finalize the mapping by selecting and renaming necessary columns
mapped_data <- mapped_data %>%
  select(question = question.y, item = item.x, variable, domain, reverse_scaled)

# Prepare the header rows
header_rows <- mapped_data %>%
  select(variable, question, item) %>%
  column_to_rownames("variable") %>% t() %>% as.data.frame() %>%
  mutate(across(everything(), as.character))

```

```

data_rows <- survey_data[-1, ] %>% mutate(across(everything(), as.character))
colnames(data_rows) <- mapped_data$variable

# Bind header_rows and data_rows together & Remove the question and items rows
# 1253 - 2 = 1251 rows
survey_clean <- bind_rows(header_rows, data_rows) %>% slice(-c(1,2))
head(survey_clean, 10)

## 3. Clean Dates and Filter Cohorts

# Convert start_date and end_date from <chr> to <dtm> (Date-Time)
survey_clean <- survey_clean %>%
  mutate(
    start_date = parse_date_time(start_date, orders = c("ymd HMS", "ymd HM", "ymd")),
    end_date = parse_date_time(end_date, orders = c("ymd HMS", "ymd HM", "ymd"))) %>%

# Remove the observations before 2022-05-27 and on 2022-08-08
filter(as.Date(start_date) >= as.Date("2022-05-27"),
as.Date(start_date) != as.Date("2022-08-08")) %>%

# Remove the empty columns: email, first_name, last_name, custom_data
select(-c("email", "first_name", "last_name", "custom_data"))

dim(survey_clean)
# 1251 - 4 (obs before May 27, 2022) - 2 (observations on August 8, 2022) = 1245 rows
# 76 - 4 (empty columns) = 72 columns

## 4. Address Typos
survey_clean <- survey_clean %>%
  mutate(across(
    c(starts_with("FOS"), starts_with("OCRBS"), starts_with("PSTLS")),

```

```

    ~str_replace_all(.x, "Neither agree (or|not|nor) disagree", "Neither agree nor disagree")
  ))

## 5. Recode Likert Scales and Demographics

# Collects unique response across all scale columns
survey_clean %>%
  select(matches("^(FOS|OCRBS|PSTLS|RFQ)")) %>%
  pivot_longer(everything(),
    names_to = "variable",
    values_to = "response") %>%
  # Create scale column by removing the numbers (e.g., FOS1 becomes FOS)
  mutate(scale = str_extract(variable, "[A-Z]+")) %>%
  # Find unique responses per scale
  distinct(scale, response) %>%
  arrange(scale, response) %>%
  print(n = 30)

# Convert all scale variables to factors and reorder their levels
survey_clean <- survey_clean %>%
  mutate(
    across(c(starts_with("FOS"), starts_with("OCRBS"), starts_with("PSTLS")),
      ~ factor(.x, levels = c("Strongly disagree",
        "Disagree",
        "Neither agree nor disagree",
        "Agree",
        "Strongly agree",
        "Prefer not to answer"), ordered = TRUE)),
    across(starts_with("RFQ"),
      ~ factor(.x, levels = c("Never",
        "Seldom",
        "Sometimes",

```

```

        "Often",
        "Very often",
        "Prefer not to answer"), ordered = TRUE))
)

# Rename the level of week_day from "1 day\tpers week" to "1 day per week"
survey_clean$work_day[survey_clean$work_day == "1 day\tpers week"] <- "1 day per week"

# Convert these variables to factor
survey_clean <- survey_clean %>%
  mutate(
    # 1. practice_year
    practice_year = factor(practice_year, levels = c(
      "Less than 1 year",
      "1-5 years",
      "6-10 years",
      "11-15 years",
      "16+ years"
    )),

    # 2.international_graduate
    international_graduate = factor(international_graduate, levels = c(
      "No",
      "Yes: please specify the country from which you graduated"
    )),

    # 3. work_day
    work_day = factor(work_day, levels = c(
      "Less than 1 day per week",
      "1 day per week",
      "2 days per week",
      "3 days per week",
      "4 days per week",

```



```

    "5 or more days per week"
  )),

# 4. practice_model
practice_model = factor(practice_model, levels = c(
  "Aboriginal Health Access Centres (AHAC)",
  "Community Health Centre (CHC)",
  "Comprehensive Care Model (CCM)",
  "Family Health Group (FHG)",
  "Family Health Network (FHN)",
  "Family Health Organization (FHO)",
  "Family Health Team (FHT)",
  "Fee-for-service (FFS)",
  "Nurse Practitioner Led Clinics (NPLC)",
  "Other (please specify)"
)),

# 5. community_size
community_size = factor(community_size, levels = c(
  "Rural (less 1,000 people)",
  "Small population center (1,000 to 29,999 people)",
  "Medium population center (30,000 to 99,999 people)",
  "Large population center (100,000 to 999,999 people)",
  "Urban centers (1 million people and over)"
)),

# 6. staff_number
staff_number = factor(staff_number, levels = c(
  "Just me", "2-5", "6-10", "11-15", "16+"
)),

# 7. gender
gender = factor(gender, levels = c(

```

```

      "Woman",
      "Man",
      "Gender neutral",
      "Intersex",
      "Non-binary",
      "Prefer not to say",
      "Prefer to self-identify:"
    ))
  )

# Select observations before Aug 4th, 2022 (cohort 1) and after Aug 8th, 2022 (cohort 3)
survey_clean <- survey_clean %>%
  filter(as.Date(start_date) < as.Date("2022-08-04") |
         as.Date(start_date) >= as.Date("2022-08-09"))
dim(survey_clean) # 225 observations

## 6. Explore Missingness & Remove NAs
skim(survey_clean)

# Heatmap
survey_clean %>%
  select(FOS1:RFQ11) %>%
  vis_miss() +
  theme(axis.text.x = element_text(angle = 90))

# Upset plot
gg_miss_upset(
  survey_clean %>% select(FOS1:RFQ11),
  nsets = n_var_miss(survey_clean %>%
                     select(FOS1:RFQ11)))

# Summarize observations with NAs across all scales

```

```

survey_clean <- survey_clean %>%
  mutate(any_NA = if_else(if_all(FOS1:RFQ11, is.na),
                           "Yes", "No"))

table1(~ any_NA, data = survey_clean,
       caption = "Observations with NAs in all scales")

# Remove observations with NAs in all scales
survey_clean <- survey_clean %>%
  filter(any_NA == "No")

# Check number of observations
dim(survey_clean)
# 206 observations

# Save cleaned data for the next steps
dir.create(here("data", "processed"), showWarnings = FALSE)
saveRDS(survey_clean, here("data", "processed", "survey_cleaned.rds"))
print("Data cleaning complete. Saved to data/processed/survey_cleaned.rds")

### 02 Descriptive Statistics
library(tidyverse)
library(here)
library(psych)
library(ggcorrplot)

## 1. Load cleaned data
survey_clean <- readRDS(here("data", "processed", "survey_cleaned.rds"))

# Create output directories if they don't exist
dir.create(here("outputs"), recursive = TRUE, showWarnings = FALSE)

```

```
## 2. Convert Scale Categories into Numbers (1-5)
```

```
survey_scale <- survey_clean %>%
```

```
  select(respondent_id, FOS1:RFQ11)
```

```
# Create table for scale items
```

```
likert_scale <- survey_clean %>%
```

```
  select(respondent_id, FOS1:RFQ11) %>%
```

```
  mutate(
```

```
    # 1. FOS, OCRBS, and PSTLS scales
```

```
    across(
```

```
      starts_with(c("FOS", "OCRBS", "PSTLS")),
```

```
      ~ case_match(
```

```
        as.character(.x),
```

```
        "Strongly disagree" ~ 1,
```

```
        "Disagree" ~ 2,
```

```
        "Neither agree nor disagree" ~ 3,
```

```
        "Agree" ~ 4,
```

```
        "Strongly agree" ~ 5,
```

```
        "Prefer not to answer" ~ NA,
```

```
        .default = NA
```

```
      )
```

```
    ),
```

```
# 2. RFQ scales
```

```
    across(
```

```
      starts_with("RFQ"),
```

```
      ~ case_match(
```

```
        as.character(.x),
```

```
        "Never" ~ 1,
```

```
        "Seldom" ~ 2,
```

```
        "Sometimes" ~ 3,
```

```
        "Often" ~ 4,
```

```

    "Very often" ~ 5,
    "Prefer not to answer" ~ NA,
    .default = NA
  )
)
) %>%
rename_all(~paste0(., "_score"))

# Combine likert scale with scale dataset
survey_scale <- cbind(survey_scale, likert_scale) %>%
  select(-respondent_id_score)

## 3. Recode the Reverse-Scored Items
reverse_likert_scale <- survey_clean %>%
  select(respondent_id, PSTLS1, PSTLS12, PSTLS15,
         RFQ1, RFQ2, RFQ4, RFQ6, RFQ8, RFQ9, RFQ11) %>%
  mutate(
    # 1. FOS, OCRBS, and PSTLS scales
    across(
      starts_with("PSTLS"),
      ~ case_match(
        as.character(.x),
        "Strongly disagree" ~ 5,
        "Disagree" ~ 4,
        "Neither agree nor disagree" ~ 3,
        "Agree" ~ 2,
        "Strongly agree" ~ 1,
        "Prefer not to answer" ~ NA
      )
    ),

    # 2. RFQ scales

```

```

    across(
      starts_with("RFQ"),
      ~ case_match(
        as.character(.x),
        "Never" ~ 5,
        "Seldom" ~ 4,
        "Sometimes" ~ 3,
        "Often" ~ 2,
        "Very often" ~ 1,
        "Prefer not to answer" ~ NA
      )
    )
  ) %>%
  rename_all(~paste0(., "_reverse_score"))

# Combine reverse likert scale with scale dataset
survey_scale <- cbind(survey_scale, reverse_likert_scale) %>%
  select(-respondent_id_reverse_score)

## 4. Organize Demographics Data
# Skim data (initial data inspection)
survey_clean %>%
  select(gender, international_graduate, practice_year,
         work_day, practice_model, community_size, staff_number) %>%
  skimr::skim()

# Combined Small Cells Categorical Data
survey_clean <- survey_clean %>%
  mutate(
    gender_merg = factor(if_else(gender %in% c("Man", "Woman"), gender, NA),
                        levels = c("Man", "Woman")),

```

```

international_graduate_merg = factor(if_else(
  international_graduate == "Yes: please specify the country from which you graduated",
  "Yes", international_graduate), levels = c("No", "Yes")),

practice_year_merg = factor(case_when(
  practice_year %in% c("Less than 1 year", "1-5 years") ~ "5 years or less",
  practice_year %in% c("6-10 years", "11-15 years", "16+ years") ~ "6 years or more"),
  levels = c("5 years or less", "6 years or more")),

work_day_merg = factor(case_when(
  work_day %in% c("Less than 1 day per week", "1 day per week",
    "2 days per week", "3 days per week") ~ "3 or fewer days per week",
  work_day %in% c("4 days per week", "5 or more days per week") ~ "4 or more days per week"),
  levels = c("3 or fewer days per week", "4 or more days per week")),

practice_model_merg = factor(if_else(
  practice_model == "Family Health Organization (FHO)", "FHO", "Other"),
  levels = c("FHO", "Other")),

community_size_merg = factor(if_else(
  community_size == "Urban centers (1 million people and over)",
  "Urban centre", "Smaller areas"), levels = c("Urban centre", "Smaller areas")),

staff_number_merg = factor(case_when(
  staff_number %in% c("Just me", "2-5") ~ "5 or less",
  staff_number %in% c("6-10", "11-15", "16+") ~ "6 or more"
), levels = c("5 or less", "6 or more"))
)

# Skim data (initial data inspection)
survey_clean %>%
  select(gender_merg, international_graduate_merg, practice_year_merg,
    work_day_merg, practice_model_merg, community_size_merg, staff_number_merg) %>%

```

```

skimr::skim()

## 5. Explore Scale Reliability (Cronbach's alpha)

# FOS utility
utility_items <- paste0("FOS", 1:5, "_score")
survey_scale %>%
  select(all_of(utility_items)) %>%
  alpha(title = "FOS utility")

# FOS accountability
accountability_items <- paste0("FOS", 6:10, "_score")
survey_scale %>%
  select(all_of(accountability_items)) %>%
  alpha(title = "FOS accountability")

# FOS self-efficacy
efficacy_items <- paste0("FOS", 11:15, "_score")
survey_scale %>%
  select(all_of(efficacy_items)) %>%
  alpha(title = "FOS self-efficacy")

# OCRBS valence
valence_items <- paste0("OCRBS", c(1, 4, 7, 10), "_score")
survey_scale %>%
  select(all_of(valence_items)) %>%
  alpha(title = "OCRBS valence")

# OCRBS discrepancy
discrepancy_items <- paste0("OCRBS", c(3, 6, 9, 12), "_score")
survey_scale %>%
  select(all_of(discrepancy_items)) %>%
  alpha(title = "OCRBS discrepancy")

```



```

## 6. Categorical Score Distribution for each Items

## Define Items
utility_items <- paste0("FOS", 1:5, "_score")
accountability_items <- paste0("FOS", 6:10, "_score")
efficacy_items <- paste0("FOS", 11:15, "_score")
valence_items <- paste0("OCRBS", c(1, 4, 7, 10), "_score")
discrepancy_items <- paste0("OCRBS", c(3, 6, 9, 12), "_score")
all_constructs <- c(utility_items, accountability_items,
  efficacy_items, valence_items, discrepancy_items)

# Print Alphas
print(alpha(survey_scale %>% select(all_of(utility_items)), title = "FOS utility"))
print(alpha(survey_scale %>% select(all_of(accountability_items)), title = "FOS accountability"))
print(alpha(survey_scale %>% select(all_of(efficacy_items)), title = "FOS self-efficacy"))
print(alpha(survey_scale %>% select(all_of(valence_items)), title = "OCRBS valence"))
print(alpha(survey_scale %>% select(all_of(discrepancy_items)), title = "OCRBS discrepancy"))

# Polychoric Correlation Matrix
item_data_wide <- survey_scale %>% select(all_of(all_constructs))
poly_res <- polychoric(item_data_wide, na.rm = TRUE)
poly_matrix <- poly_res$rho

p_corr <- ggcorrplot(poly_matrix, type = "full", outline.col = "white",
  colors = c("#6D9EC1", "white", "#E46726"),
  title = "Item-Level Polychoric Correlations", ggtheme = theme_minimal())

dir.create(here("outputs", "figures"), recursive = TRUE, showWarnings = FALSE)
ggsave(here("outputs", "figures", "polychoric_correlation.png"),
  plot = p_corr, width = 10, height = 10)

# Save lower matrix to CSV

```

```

lower_matrix <- round(poly_matrix, 2)
lower_matrix[upper.tri(lower_matrix, diag = TRUE)] <- ""
write.csv(as.data.frame(lower_matrix), here("outputs", "polychoric_lower_matrix.csv"))

## 7. Merge scores with demographics and save
survey_clean <- survey_clean %>% mutate(rownum = row_number())
survey_scale <- survey_scale %>% mutate(rownum = row_number())

survey_final_export <- left_join(
  survey_clean,
  survey_scale %>% select(rownum, ends_with("score")),
  by = "rownum"
) %>% select(-rownum)

saveRDS(survey_final_export, here("data", "processed", "survey_scored.rds"))

### 03 CFA Modeling
library(tidyverse)
library(here)
library(lavaan)
library(psych)
library(kableExtra)

survey_clean <- readRDS(here("data", "processed", "survey_scored.rds"))

# Create output directories if they don't exist
dir.create(here("outputs", "figures"), recursive = TRUE, showWarnings = FALSE)

## 1. Define variables as ordered for WLSMV

utility_items <- paste0("FOS", 1:5, "_score")

```

```

accountability_items <- paste0("FOS", 6:10, "_score")
efficacy_items <- paste0("FOS", 11:15, "_score")
valence_items <- paste0("OCRBS", c(1, 4, 7, 10), "_score")
discrepancy_items <- paste0("OCRBS", c(3, 6, 9, 12), "_score")

all_constructs <- c(utility_items, accountability_items, efficacy_items,
                    valence_items, discrepancy_items)

cfa_data_raw <- survey_clean %>%
  select(all_of(all_constructs)) %>%
  mutate(across(everything(), as.ordered))

## 2. Fit 5-Factor Model
m5f_syntax <- paste0(
  'Utility      =~ ', paste(utility_items, collapse = " + "), '\n',
  'Accountability =~ ', paste(accountability_items, collapse = " + "), '\n',
  'Self_Efficacy =~ ', paste(efficacy_items, collapse = " + "), '\n',
  'Valence       =~ ', paste(valence_items, collapse = " + "), '\n',
  'Discrepancy   =~ ', paste(discrepancy_items, collapse = " + ")
)

fit5f_pairwise <- cfa(model = m5f_syntax, data = cfa_data_raw,
                      estimator = "WLSMV", missing = "pairwise")

# Extract validated factor scores
latent_scores <- lavPredict(fit5f_pairwise)
used_indices <- lavInspect(fit5f_pairwise, "case.idx")[[1]]

# Create score table with indices
factor_scores <- as.data.frame(latent_scores)
survey_clean_final <- bind_cols(survey_clean, factor_scores)

# Final Fit Summary

```

```

summary(fit5f_pairwise, fit.measures = TRUE, standardized = TRUE)

# Latent Correlations
latent_matrix <- lavInspect(fit5f_pairwise, what = "cor.lv")
print(latent_matrix)

## 3. Fit Comparison Models (4F, 2F, 1F)

# (1) 4-Factor Model: Merging Utility and Valence
# Tests if "Utility" and "Valence" are actually a single value construct
m4f_syntax <- paste0(
  'Util_Val_Merged =~ ', paste(c(utility_items, valence_items), collapse = " + "), '\n',
  'Accountability =~ ', paste(accountability_items, collapse = " + "), '\n',
  'Self_Efficacy =~ ', paste(efficacy_items, collapse = " + "), '\n',
  'Discrepancy =~ ', paste(discrepancy_items, collapse = " + ")
)

# Fit 4-Factor Model
fit4f <- cfa(
  model = m4f_syntax,
  data = cfa_data_raw,
  estimator = "WLSMV",
  missing = "pairwise"
)

# Compare against the 5-factor model
comp_5v4 <- lavTestLRT(fit5f_pairwise, fit4f)

print("Comparison: 5-Factor vs. 4-Factor (Merging Utility and Valence)")
print(comp_5v4)

# (2) 4-Factor Model: Merging Utility and Accountability

```

```

# Tests if "Utility" and "Accountability" are actually a single value construct
# because these two have the highest correlation (0.86)
m4f_util_acc_syntax <- paste0(
  'Util_Acc_Merged =~ ', paste(c(utility_items, accountability_items), collapse = " + "), '\n',
  'Self_Efficacy =~ ', paste(efficacy_items, collapse = " + "), '\n',
  'Valence =~ ', paste(valence_items, collapse = " + "), '\n',
  'Discrepancy =~ ', paste(discrepancy_items, collapse = " + ")
)

# Fit 4-factor the model
fit4f_util_acc <- cfa(
  model = m4f_util_acc_syntax,
  data = cfa_data_raw,
  estimator = "WLSMV",
  missing = "pairwise"
)

# Compare against the 5-factor model
comp_5v4_util_acc <- lavTestLRT(fit5f_pairwise, fit4f_util_acc)

print("Comparison: 5-Factor vs. 4-Factor (Merging Utility and Accountability)")
print(comp_5v4_util_acc)

## (3) 2-Factor Model: FOS vs. OCRBS
# Tests if items simply group by the scales they came from (FOS vs. OCRBS)
m2f_syntax <- paste0(
  'All_FOS =~ ', paste(c(utility_items, accountability_items,
                        efficacy_items), collapse = " + "), '\n',
  'All_OCRBS =~ ', paste(c(valence_items, discrepancy_items), collapse = " + ")
)

# Fit 2-Factor Model
fit2f <- cfa(

```

```

model      = m2f_syntax,
data       = cfa_data_raw,
estimator  = "WLSMV",
missing    = "pairwise"
)

# Compare 5-Factor vs. 2-Factor
comp_5v2 <- lavTestLRT(fit5f_pairwise, fit2f)

print("Comparison: 5-Factor vs. 2-Factor (FOS vs. OCRBS)")
print(comp_5v2)

## (4) 1-Factor Model: Merge All Constructs
# Tests if all items simply measure one general dimension
m1f_syntax <- paste0(
  'Global_Factor =~ ', paste(all_constructs, collapse = " + ")
)

# Fit 1-Factor Model
fit1f <- cfa(
  model = m1f_syntax,
  data = cfa_data_raw,
  estimator = "WLSMV",
  missing = "pairwise"
)

# Compare 5-Factor vs. 1-Factor
comp_5v1 <- lavTestLRT(fit5f_pairwise, fit1f)

print("Comparison: 5-Factor vs. 1-Factor (Global)")
print(comp_5v1)

## 4. Create CFA Model Comparison Table

```

```

# Standardizing the Fit Indices Function

get_fit_indices <- function(model_fit, model_name) {
  indices <- fitMeasures(model_fit, c("chisq.scaled", "df.scaled", "pvalue.scaled",
                                     "cfi.scaled", "tli.scaled",
                                     "cfi.robust", "tli.robust",
                                     "rmsea.scaled", "srmr"))

  p_val <- indices["pvalue.scaled"]
  p_txt <- ifelse(p_val < 0.001, "<b>< .001</b>", sprintf("%.3f", p_val))

  data.frame(
    Model = model_name,
    Chi_Sq = round(indices["chisq.scaled"], 2),
    df = as.integer(indices["df.scaled"]),
    p = p_txt,
    CFI = sprintf("%.3f", indices["cfi.scaled"]),
    TLI = sprintf("%.3f", indices["tli.scaled"]),
    CFI_Robust = sprintf("%.3f", indices["cfi.robust"]),
    TLI_Robust = sprintf("%.3f", indices["tli.robust"]),
    RMSEA = sprintf("%.3f", indices["rmsea.scaled"]),
    SRMR = sprintf("%.3f", indices["srmr"])
  )
}

# Creating the Table Object
comparison_table <- bind_rows(
  get_fit_indices(fit5f_pairwise, "5-Factor (Hypothesized)"),
  get_fit_indices(fit4f, "4-Factor (Merge Utility/Valence)"),
  get_fit_indices(fit4f_util_acc, "4-Factor (Merge Utility/Accountability)"),
  get_fit_indices(fit2f, "2-Factor (FOS vs OCRBS)"),
  get_fit_indices(fit1f, "1-Factor (Global)")
)

```

```

# Generate and Save Table
comparison_table %>%
  rename(
    "&chi;<sup>2</sup> (scaled)" = Chi_Sq,
    "<i>df</i>" = df,
    "<i>p</i>" = p,
    "Robust CFI" = CFI_Robust,
    "Robust TLI" = TLI_Robust
  ) %>%
  kable(
    format = "html",
    escape = FALSE,
    align = "lcccccccc",
    row.names = FALSE,
    caption = "<b>Table 1: Comparison of CFA Model Fit Indices (N=206)</b>"
  ) %>%
  kable_styling(
    bootstrap_options = c("striped", "hover", "condensed"),
    full_width = FALSE,
    font_size = 16
  ) %>%
  add_header_above(c(" " = 1, "Absolute Fit" = 3,
                     "Incremental Fit" = 4,
                     "Residual Fit" = 2)) %>%
  row_spec(1, bold = TRUE, background = "#f8fafc") %>%
  footnote(
    general = "Estimator: WLSMV with polychoric correlations to
    respect the ordinal nature of 5-point Likert items.",
    general_title = "Note: ",
    footnote_as_chunk = TRUE
  )
comparison_table

```



```

comparison_table %>% save_kable(file = "outputs/figures/CFA_Model_Comparison.png")

saveRDS(survey_clean_final, here("data", "processed", "survey_cfa_scores.rds"))

### 04 LPA Analysis

library(tidyverse)
library(here)
library(tidyLPA)
library(kableExtra)
library(gt)

survey_clean_final <- readRDS(here("data", "processed", "survey_cfa_scores.rds"))

# Create output directories if they don't exist
dir.create(here("outputs", "figures"), recursive = TRUE, showWarnings = FALSE)
dir.create(here("outputs", "tables"), recursive = TRUE, showWarnings = FALSE)

## 1. Fit Complex LPA Model for Comparison (1-6 profiles, multiple models)

lpa_model <- survey_clean_final %>%
  select(Utility, Self_Efficacy, Valence, Discrepancy) %>%
  estimate_profiles(n_profiles = 1:6,
                   models = c(1, 2, 3, 6))

# Create, display, and save the Model Comparison Table
model_comparison <- get_fit(lpa_model) %>%
  arrange(BIC) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  mutate(
    BLRT_p = cell_spec(BLRT_p, color = ifelse(is.na(BLRT_p) | BLRT_p > 0.05,

```

```

                                "red", "inherit")),
n_min = cell_spec(n_min, color = ifelse(is.na(n_min) | n_min < 0.05,
                                "red", "inherit")),
Entropy = cell_spec(Entropy, color = ifelse(is.na(Entropy) | Entropy < 0.8,
                                "red", "inherit"))
) %>%
kable(
  format = "html",
  align = "c",
  caption = "Complex LPA Model",
  escape = FALSE
) %>%
collapse_rows(columns = 1, valign = "top") %>%
kable_styling(
  bootstrap_options = c("striped", "hover"),
  font_size = 12,
  full_width = FALSE
)

print(model_comparison)

save_kable(model_comparison, file = here("outputs", "figures",
  "LPA_Model_Comparison.png"), vwidth = 2000)

compare_solutions(lpa_model)

# Show and save the individual profiles plot
p_profiles <- plot_profiles(lpa_model)
print(p_profiles)
ggsave(here("outputs", "figures", "LPA_plot_profiles.png"),
  plot = p_profiles, width = 18, height = 10)

```

```

## 2. Fit Candidate LPA Models

# (1) 3-Class, Model 1
final_lpa_c3m1 <- survey_clean_final %>%
  select(Utility, Self_Efficacy, Valence, Discrepancy) %>%
  estimate_profiles(n_profiles = 3, models = 1)
get_fit(final_lpa_c3m1)

# (2) 4-Class, Model 6
final_lpa_c4m6 <- survey_clean_final %>%
  select(Utility, Self_Efficacy, Valence, Discrepancy) %>%
  estimate_profiles(n_profiles = 4, models = 6)
get_fit(final_lpa_c4m6)

# (3) 2-Class, Model 3
final_lpa <- survey_clean_final %>%
  select(Utility, Self_Efficacy, Valence, Discrepancy) %>%
  estimate_profiles(n_profiles = 2, models = 3)
get_fit(final_lpa)
lpa_results <- get_data(final_lpa)

## 3. LPA Candidate Model Comparison Table
model_comp_clean <- get_fit(lpa_model) %>%
  filter((Classes == 2 & Model == 3) |
         (Classes == 3 & Model == 1) |
         (Classes == 4 & Model == 6)) %>%
  mutate(
    Model_Name = case_when(
      Classes == 2 & Model == 3 ~ "2-Class (Model 3)",
      Classes == 3 & Model == 1 ~ "3-Class (Model 1)",
      Classes == 4 & Model == 6 ~ "4-Class (Model 6)"
    ),
    across(c(LogLik, AIC, BIC, SABIC, Entropy), ~ round(as.numeric(.x), 2)),

```

```

    n_min = sprintf("%.1f%%", n_min * 100),
    BLRT_p = ifelse(BLRT_p < 0.001, "< 0.001", sprintf("%.3f", as.numeric(BLRT_p)))
  ) %>%
select(Model_Name, parameters, LogLik, AIC, BIC, SABIC, Entropy, n_min, BLRT_p)

# Generate comparison table
comparison_table <- model_comp_clean %>%
  gt() %>%
  tab_header(
    title = md("**Comparison of Candidate Latent Profile Models**"),
    subtitle = md("*Fit Indices and Class Size Characteristics*")
  ) %>%
  cols_label(
    Model_Name = md("**Model**"),
    parameters = md("**Parameters**"),
    LogLik = md("**Log-Likelihood**"),
    AIC = md("**AIC**"),
    BIC = md("**BIC**"),
    SABIC = md("**SABIC**"),
    Entropy = md("**Entropy**"),
    n_min = md("**Smallest Class (%)**"),
    BLRT_p = md("**BLRT *p*-value**")
  ) %>%
  cols_align(aligned = "center", columns = everything()) %>%
  cols_align(aligned = "left", columns = c(Model_Name)) %>%
  opt_row_stripping() %>%
  tab_options(
    heading.align = "center",
    table.border.top.style = "solid",
    table.border.top.width = px(2),
    table.border.bottom.style = "solid",
    table.border.bottom.width = px(2),
    column_labels.border.bottom.style = "solid",

```

```

column_labels.border.bottom.width = px(2),
table_body.border.bottom.style = "solid",
table_body.border.bottom.width = px(1),
data_row.padding = px(3),
table.font.size = px(13)
) %>%
tab_source_note(
  source_note = md("Note: AIC = Akaike Information Criterion;
                    BIC = Bayesian Information Criterion; SABIC = Sample-Size Adjusted BIC;
                    BLRT = Bootstrapped Likelihood Ratio Test.")
)

# Display the table
print(comparison_table)

# Save the table
comparison_table %>%
  gt_save(here("outputs", "tables", "LPA_Candidate_Models_Comparison.html"))

# The final decision is 2-Class Model 3 based on BIC
# Print Summary Table for Final Model (2-Class Model 3)
summary_table <- lpa_results %>%
  mutate(Profile = factor(Class, levels = c(1, 2))) %>%
  group_by(Profile) %>%
  summarise(
    N = n(),
    'Pct(%)' = n()/206*100,
    Utility = sprintf("%.2f (%.2f)", mean(Utility), sd(Utility)),
    Self_Efficacy = sprintf("%.2f (%.2f)", mean(Self_Efficacy), sd(Self_Efficacy)),
    Valence = sprintf("%.2f (%.2f)", mean(Valence), sd(Valence)),
    Discrepancy = sprintf("%.2f (%.2f)", mean(Discrepancy), sd(Discrepancy)),
    .groups = 'drop'
  )

```

```

)

summary_table %>%
  kable(
    format = "html",
    align = "c",
    caption = "Final LPA Results (2-Class Model 3)"
  ) %>%
  kable_styling(
    bootstrap_options = c("striped", "hover"),
    font_size = 12,
    full_width = FALSE
  ) %>%
  column_spec(1, bold = TRUE) %>%
  footnote(general = "Mean (SD)") %>%
  scroll_box(width = "100%")

## 3. Prepare Data for Regression and Plotting
regression_data <- survey_clean_final %>%
  mutate(
    Class = factor(lpa_results$Class),
    Profile = factor(lpa_results$Class,
                     labels = c("Low-Appraisal", "High-Appraisal"))
  )

profile_summary <- regression_data %>%
  group_by(Profile) %>%
  summarise(n = n(), .groups = "drop") %>%
  mutate(pct = round((n/sum(n)) * 100, 1),
         facet_label = paste0(Profile, " (n = ", n, ", ", pct, "%)"))

plot_data <- lpa_results %>%

```

```

mutate(Profile = factor(Class, labels = c("Low-Appraisal", "High-Appraisal"))) %>%
left_join(profile_summary, by = "Profile") %>%
group_by(Profile, facet_label) %>%
summarise(
  Utility = mean(Utility),
  Self_Efficacy = mean(Self_Efficacy),
  Valence = mean(Valence),
  Discrepancy = mean(Discrepancy),
  .groups = 'drop'
) %>%
pivot_longer(cols = c(Utility, Self_Efficacy, Valence, Discrepancy),
             names_to = "Construct",
             values_to = "Mean_Score") %>%
mutate(Construct = factor(Construct, levels = c("Utility",
        "Self_Efficacy", "Valence", "Discrepancy")))

## 4. Visualizing Final LPA Results

# Individual Assignment Probabilities Plot
prob_data <- lpa_results %>%
  mutate(
    Max_Prob = pmax(CPROB1, CPROB2),
    Profile = factor(Class, labels = c("Low-Appraisal", "High-Appraisal"))
  )

p_prob <- ggplot(prob_data, aes(x = Profile, y = Max_Prob, color = Profile)) +
  geom_jitter(width = 0.2, size = 3, alpha = 0.6) +
  scale_color_manual(values = c("High-Appraisal" = "#2ca02c",
                                "Low-Appraisal" = "#d62728")) +
  scale_y_continuous(limits = c(0, 1), labels = scales::percent_format()) +
  labs(
    title = "Individual Assignment Probabilities by Latent Profile (N = 206)",

```

```

    subtitle = "Each dot is one physician.",
    x = "Assigned Profile",
    y = "Probability of Belonging to Assigned Profile"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 12, hjust = 0),
    axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1, size = 10),
    axis.text.y = element_text(size = 10),
    legend.title = element_text(face = "bold")
  )

print(p_prob)

ggsave(here("outputs", "figures",
           "LPA_individual_probabilities.png"), plot = p_prob, width = 14, height = 6)

# Dot and Line Plot (Mean Scores)
p_means <- ggplot(plot_data, aes(x = Construct, y = Mean_Score, color = Profile, group = Profile)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray60", linewidth = 1) +
  geom_line(linewidth = 1.5) +
  geom_point(size = 4) +
  scale_color_manual(values = c("High-Appraisal" = "#2ca02c",
                                "Low-Appraisal" = "#d62728"),
                     labels = unique(plot_data$facet_label)) +
  labs(
    title = "Comparing Physician Belief Profiles",
    subtitle = "Dashed line represents the overall sample average.",
    x = "Construct",
    y = "Standardized Score",
    color = "Profile"
  ) +
  theme_minimal() +

```



```

theme(
  plot.title = element_text(face = "bold", size = 12, hjust = 0),
  axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1, size = 10),
  axis.text.y = element_text(size = 10),
  legend.title = element_text(face = "bold")
)

print(p_means)
ggsave(here("outputs", "figures", "LPA_mean_scores.png"),
  plot = p_means, width = 10, height = 6)

# Scatter Plot: Utility vs Valence (with ellipses and centroids)
centroids <- regression_data %>%
  group_by(Profile) %>%
  summarise(Utility_Center = mean(Utility), Valence_Center = mean(Valence), .groups = 'drop')

p_cluster <- ggplot(regression_data, aes(x = Utility, y = Valence, color = Profile)) +
  geom_point(alpha = 0.5, size = 2.5) +
  stat_ellipse(level = 0.95, linetype = "dashed", linewidth = 0.8) +
  geom_point(data = centroids, aes(x = Utility_Center, y = Valence_Center, fill = Profile),
    shape = 23, size = 6, color = "black", stroke = 1.5) +
  scale_color_manual(values = c("High-Appraisal" = "#2ca02c", "Low-Appraisal" = "#d62728")) +
  scale_fill_manual(values = c("High-Appraisal" = "#2ca02c", "Low-Appraisal" = "#d62728")) +
  theme_minimal() +
  labs(title = "Cluster Distribution: Utility vs. Valence",
    x = "Utility (Standardized Score)",
    y = "Valence (Standardized Score)")

print(p_cluster)
ggsave(here("outputs", "figures", "cluster_utility_valence.png"),
  plot = p_cluster, width = 10, height = 6)

```

```

## 5. Save Final Data for Regression
saveRDS(regression_data, here("data", "processed", "survey_lpa_assigned.rds"))

### 05 Regression Analysis

library(tidyverse)
library(here)
library(gtsummary)
library(car)
library(sjPlot)

# Create output directories if they don't exist
dir.create(here("outputs", "figures"), recursive = TRUE, showWarnings = FALSE)
dir.create(here("outputs", "tables"), recursive = TRUE, showWarnings = FALSE)

# Load data
regression_data <- readRDS(here("data", "processed", "survey_lpa_assigned.rds"))

## 1. Data Preparation
regression_data_complete <- regression_data %>%
  drop_na(Profile, gender_merg, international_graduate_merg, practice_year_merg,
          work_day_merg, community_size_merg, staff_number_merg, practice_model_merg)

# Generate a descriptive statistics table
desc_table <- regression_data_complete %>%
  select(
    Profile,
    gender_merg,
    international_graduate_merg,
    practice_year_merg,

```

```

    work_day_merg,
    community_size_merg,
    staff_number_merg,
    practice_model_merg
  ) %>%
tbl_summary(
  by = Profile,
  label = list(
    gender_merg ~ "Gender",
    international_graduate_merg ~ "International Graduate",
    practice_year_merg ~ "Years in Practice",
    work_day_merg ~ "Work Days per Week",
    community_size_merg ~ "Practice Community Size",
    staff_number_merg ~ "Number of Staff",
    practice_model_merg ~ "Practice Model"
  ),
  statistic = list(all_categorical() ~ "{n} ({p}%)",
    missing = "no"
  ) %>%
add_overall()

# Print the table
desc_table

# Save descriptive statistics table
desc_table %>%
  as_gt() %>%
  gt::gtsave(here("outputs", "tables", "descriptive_statistics.html"))

## 2. Fit Initial Models

# Simple Linear Regression

```

```

simple_model <- lm(Accountability ~ Profile, data = regression_data_complete)
print("--- Simple Model Summary ---")
summary(simple_model)

# Multiple Linear Regression
full_model <- lm(Accountability ~ Profile + gender_merg + international_graduate_merg +
                  practice_year_merg + work_day_merg + community_size_merg + staff_number_merg +
                  practice_model_merg, data = regression_data_complete)
print("--- Full Model Summary ---")
summary(full_model)

## 3. Model Selection
n_obs <- nobs(simple_model)

# Stepwise BIC
stepwise_bic <- step(
  simple_model,
  direction = "backward",
  scope = list(lower = simple_model, upper = full_model),
  k = log(n_obs),
  trace = FALSE
)
print("--- Stepwise BIC Selection ---")
summary(stepwise_bic)

# Stepwise AIC
stepwise_aic <- step(
  simple_model,
  direction = "both",
  scope = list(lower = simple_model, upper = full_model),
  trace = FALSE
)

```

```

print("--- Stepwise AIC Selection ---")
summary(stepwise_aic)

## 4. Final Linear Model
# Select the final model based on the bidirectional stepwise AIC results:
final_model <- lm(Accountability ~ Profile + gender_merg +
  international_graduate_merg + work_day_merg, data = regression_data_complete)
print("--- Final Model Summary ---")
summary(final_model)

## 5. Model Validation (Diagnostics)
# Check Multicollinearity
print("--- Variance Inflation Factor (VIF) ---")
print(vif(final_model))

# Save 4-panel diagnostic plot for Linearity, Normality, Homoscedasticity, Outliers
png(here("outputs", "figures", "regression_diagnostics.png"), width = 10,
  height = 8, units = "in", res = 300)
par(mfrow = c(2, 2))
plot(final_model)
par(mfrow = c(1, 1))
dev.off()

## 6. Visualizing the Model
p_forest <- plot_model(
  final_model,
  type = "est",
  show.values = TRUE,
  value.offset = .3,
  title = "Predictors of Accountability",

```

```

    vline.color = "red"
  ) + theme_sjplot()

print(p_forest)

ggsave(here("outputs", "figures", "regression_forest_plot.png"),
       plot = p_forest, width = 8, height = 6)

## 7. Generate Final Output Table
# Model Comparison Table
format_reg_tbl <- function(model) {
  all_labels <- list(
    Profile = "Physician Belief Profiles",
    gender_merg = "Gender",
    international_graduate_merg = "International Graduate",
    practice_year_merg = "Years in Practice",
    work_day_merg = "Work Days per Week",
    community_size_merg = "Practice Community Size",
    staff_number_merg = "Number of Staff",
    practice_model_merg = "Practice Model"
  )
  model_vars <- names(model$model)
  used_labels <- all_labels[names(all_labels) %in% model_vars]

  tbl_regression(
    model,
    intercept = TRUE,
    pvalue_fun = ~style_pvalue(.x, digits = 3),
    estimate_fun = ~style_sigfig(.x, digits = 2),
    label = used_labels
  ) %>%
  add_global_p(keep = TRUE) %>%
  add_glance_table(

```

```

      include = c(r.squared, adj.r.squared, statistic, p.value, AIC, BIC, nobs),
      label = list(
        r.squared ~ "R2",
        adj.r.squared ~ "Adjusted R2",
        nobs ~ "Total Physicians (N)"
      )
    ) %>%
    bold_p(t = 0.05) %>%
    bold_labels() %>%
    italicize_levels()
  }

tbl_full <- format_reg_tbl(full_model)
tbl_aic <- format_reg_tbl(stepwise_aic)
tbl_bic <- format_reg_tbl(stepwise_bic)

final_comparison_table <- tbl_merge(
  tbls = list(tbl_full, tbl_aic, tbl_bic),
  tab_spanner = c("**Full Model**", "**Stepwise (AIC) / Final**", "**Stepwise (BIC)**")
) %>%
  modify_header(
    estimate_1 = "**Beta**", p.value_1 = "**p-value**",
    estimate_2 = "**Beta**", p.value_2 = "**p-value**",
    estimate_3 = "**Beta**", p.value_3 = "**p-value**"
  ) %>%
  modify_footnote(everything() ~ NA) %>%
  as_gt() %>%
  tab_header(
    title = md("**Comparison of Regression Models (2-Class Model 3)**"),
    subtitle = md("*Evaluating Variable Retention and Model Fit across Selection Criteria*")
  ) %>%
  tab_options(
    heading.align = "center",

```

```

    table.border.top.style = "solid",
    table.border.top.width = px(2),
    table.border.bottom.style = "solid",
    table.border.bottom.width = px(2),
    column_labels.border.bottom.style = "solid",
    column_labels.border.bottom.width = px(2),
    table_body.border.bottom.style = "solid",
    table_body.border.bottom.width = px(1),
    data_row.padding = px(3),
    table.font.size = px(13)
  )

final_comparison_table
final_comparison_table %>%
  gt::gtsave(here("outputs", "tables", "regression_model_comparison.html"))

# Create the base summary table
reg_table <- final_model %>%
  tbl_regression(
    intercept = FALSE,
    estimate_fun = ~style_sigfig(.x, digits = 2),
    pvalue_fun = ~style_pvalue(.x, digits = 3),
    label = list(
      Profile ~ "Physician Belief Profiles",
      gender_merg ~ "Gender",
      international_graduate_merg ~ "International Graduate",
      work_day_merg ~ "Work Days per Week"
    )
  ) %>%
  add_global_p(keep = TRUE) %>%
  bold_labels() %>%
  italicize_levels() %>%
  bold_p(t = 0.05) %>%

```



```

modify_header(
  label = "**Characteristic**",
  estimate = "**Beta (95% CI)**",
  p.value = "**p-value**"
) %>%
modify_footnote(everything() ~ NA)

# Extract model fit statistics
n_obs <- nobs(final_model)
adj_r2 <- round(summary(final_model)$adj.r.squared, 3)

final_table <- reg_table %>%
  as_gt() %>%
  tab_header(
    title = md("**Predictors of Physician Accountability**"),
    subtitle = md("*Linear Regression Adjusted for Practice and Demographic Characteristics*")
  ) %>%
  tab_source_note(source_note = sprintf("Adjusted R2 = %.3f;
    Total Physicians (N) = %d", adj_r2, n_obs)) %>%
  tab_options(
    heading.align = "center",
    table.border.top.style = "solid",
    table.border.top.width = px(2),
    table.border.bottom.style = "solid",
    table.border.bottom.width = px(2),
    column_labels.border.bottom.style = "solid",
    column_labels.border.bottom.width = px(2),
    table_body.border.bottom.style = "solid",
    table_body.border.bottom.width = px(1),
    data_row.padding = px(3)
  )
final_table

```

```
# Save table
final_table %>%
  gt::gtsave(here("outputs", "tables", "final_regression_results.html"))

print("Analysis complete. Regression table saved to outputs/tables/final_regression_results.html")

# End of Script
```

Appendix C: Artificial Intelligence (AI) Acknowledgement

I used Gemini strictly as a learning aid to check grammar, punctuation, and spelling. All mechanical suggestions were manually reviewed and selectively incorporated.